

claude-windows / d6fe19e9-affd-49ed-a202-071932fe0c3a

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\d6fe19e9-affd-49ed-a202-071932fe0c3a.jsonl`  
- SHA-256: `44510e2a04be541194c53b72b3ead4a533273288c9baf5cf1f700f6777ce1d85`  
- Source modified: `2026-06-17T14:11:29+00:00`  
- Imported at: `2026-07-05T16:48:19+00:00`  
- project: `C--Users-avidu-Projects-badminton-highlight-indexer`  
- session_id: `d6fe19e9-affd-49ed-a202-071932fe0c3a`
```

Transcript

1. user (2026-06-17T09:04:01.699Z)

Read the GitHub PR comment and incorporate it if reasonable and then merge the PR

2. assistant (2026-06-17T09:04:05.295Z)

I'll find the open PR for the current branch and read its comments.

3. user (2026-06-17T09:04:12.939Z)

```
docs/golden-regression-fixtures-plan  
---
```

4. user (2026-06-17T09:04:19.093Z)

```
198      docs(eval): R2 design ? task-faithful tolerance-match rally metric (FOR REVIEW)  
docs/r2-eval-metric      OPEN      2026-06-17T08:35:30Z  
===STATUS===
```

Relevant pull requests in avidullu/badminton-highlight-indexer

Current branch

```
#186 docs: freeze Golden Regression Fixtures plan + ... [docs/golden-regression-fixtures-plan] - Merged
```

Created by you

```
#198 docs(eval): R2 design ? task-faithful tolerance... [docs/r2-eval-metric]  
? Checks passing
```

Requesting a code review from you

You have no pull requests to review

5. assistant (2026-06-17T09:04:23.122Z)

The current branch's PR (#186) is already merged. The open PR is
[#198](https://github.com/avidullu/badminton-highlight-indexer/pull/198). Let me read its comments.

6. user (2026-06-17T09:04:26.886Z)

```
{"baseRefName":"master","body":"**For your deeper look before we build R2** (left open, not  
auto-merged).\n\nThe R2 metric design: replace `tIoU-F1@0.5` (wrong in both directions ? over-  
penalizes the fuzzy rally START, murky on over-production) with an **asymmetric tolerance-match  
under strict Hungarian 1:1**, reported as a **decomposition** (`P@?` / `R@?` / `F1@?` +  
start/end boundary-error distributions + over-seg guard + ?-sweep).\n\n**Grounded in a computed  
preview** on the two ground-truth clips ? including the honest, slightly-uncomfortable
```

findings:\n- A strict 1:1 tolerance metric makes local look **farther behind** than tIoU (it correctly charges the ~1.5x over-production + loose ends that tIoU's overlap-matching absorbed). The gap **widens** ? that's the ruler working.\n- Even **Gemini** scores low at tight ? (mp2 `tolF1@1.5` = 0.20) ? the golden **END** convention carries >1.5 s of noise ? **must** be IAA-calibrated (a small 2nd-labeler study), not hard-picked.\n- `tolF1@~3.0` ? `tIoU@0.5` ? the sweep smoothly contains tIoU as a point.\n\nIncludes: the small-n promotion rule (sign-test sweep + per-video over-seg no-regression + dual-set), the impl plan (`backend/eval/tolerance\_metrics.py`, default-additive, zero production risk), DoD, and **open** decisions for you (asymmetric vs symmetric ?_end; IAA now vs after; augment vs replace the gate).\n\n**Open** questions are in §7. Docs-only; R2 implementation is owner-gated on your review.\n\n? Generated with [Claude Code](https://claude.com/claude-code)\n", "headRefName": "docs/r2-eval-metric", "mergeable": "MERGEABLE", "reviewDecision": "", "state": "OPEN", "title": "docs(eval): R2 design ? task-faithful tolerance-match rally metric (FOR REVIEW)", "url": "https://github.com/avidullu/badminton-highlight-indexer/pull/198"}\n===COMMENTS===\nauthor: avidullu\nassociation: owner\nedited: false\nstatus: commented\n--\nI agree with both of the first recommendations but in the last one we do argument but before replacing we should run an ablation experiment and deprecate with documentation so it's reserved for posterity\n--

7. assistant (2026-06-17T09:04:33.287Z)

Let me look at the PR's actual doc changes and §7 (open decisions).

8. user (2026-06-17T09:04:34.991Z)

docs/R2_EVAL_METRIC_DESIGN.md

9. user (2026-06-17T09:04:40.847Z)

From https://github.com/avidullu/badminton-highlight-indexer
c483346..369e90c master -> origin/master
38808f0 docs(eval): R2 design ? task-faithful tolerance-match rally metric (for review)

10. user (2026-06-17T09:04:46.567Z)

?? artifacts/
?? docs/KHELSUTRA_PRODUCT_OVERVIEW.md
---clean?---

11. user (2026-06-17T09:05:08.601Z)

Exit code 128
fatal: 'docs/r2-eval-metric' is already used by worktree at 'C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/docs-r2'

12. assistant (2026-06-17T09:05:12.410Z)

The branch lives in a worktree. Let me read §7 of the doc there.

13. user (2026-06-17T09:05:33.884Z)

Already up to date.
38808f0 docs(eval): R2 design ? task-faithful tolerance-match rally metric (for review)

14. user (2026-06-17T09:05:37.026Z)

```

1      # R2 ? Task-faithful rally-boundary eval metric (tolerance-match + over-seg guard)
2
3      **Status:** **DESIGN ? for owner review (NOT started).** Gates the rally-end work (R4)
and the dual-eval-set
4      regression gate. A focused increment of the rally-quality project
([RALLY_QUALITY_RESEARCH.md](RALLY_QUALITY_RESEARCH.md) §7;
5      the day-1 validation that motivates it:
[REAL_FOOTAGE_VALIDATION_REPORT.md](REAL_FOOTAGE_VALIDATION_REPORT.md);
6      the directive: `eval-boundary-tolerance` + `eval-dual-set-regression`).
7
8      ---
9
10     ## 0. TL;DR
11     `tIoU-F1@0.5` is the wrong ruler for rally detection, in both directions: it over-
penalizes the
12     intrinsically fuzzy rally START (the ~1?1.5 s service window), and its overlap
matching is murky about
13     over-production. R2 replaces the headline with a tolerance-match metric under
strict 1:1 (Hungarian)
14     assignment, reported as a decomposition ? `P@? / R@? / F1@?` + boundary-error
distributions split
15     into start/end + an over-segmentation guard + a -sweep ? not a single number.
16
17     Honest, slightly uncomfortable consequence (measured, see §3): under R2 the local
detector's headline
18     number drops and the gap to Gemini widens vs what tIoU showed ? because R2
faithfully charges for the
19     1.5× over-production and the loose ends that tIoU's overlap matching partly absorbed.
That is the point:
20     R2 splits "behind Gemini" into its two real causes ? (a) over-production [precision]
and (b) loose
21     rally-ends [the R4 lever] ? and forgives only the inherent start fuzziness. It is a
more honest ruler,
22     not a flattering one.
23
24     ---
25
26     ## 1. Why `tIoU@0.5` is the wrong ruler (grounded in the n=2 ground-truth clips)
27     1. Over-penalizes the START. For a 5.7 s rally, `tIoU ? 0.5` tolerates only ~±1.9 s
of boundary shift
28     (`? = D·(1?)/(1+?)`). The service window (stance ? racquet contact) alone is ~1?1.5
s ? even Gemini's
29     median  $|?start|$  is 0.98?1.46 s. So tIoU spends most of its budget on a boundary that
is intrinsically
30     undefined to ~1 s. (Owner directive: don't strictly align starts; tolerate ~±2 s.)
31     2. Count parity is a deceptive proxy. mahadevpura-2 hit 1.03× of Gemini's count yet
F1@0.5 = 0.150 ?
32     right count, wrong boundaries. A count- or overlap-based view hides this.
33     3. It flips config rankings on noise. The hard-cap was golden-neutral at n=6 (+0.006
n.s.) but "best" at
34     n=7 once a continuously-active clip entered ? a metric artifact at the 0.5 cliff, not
a quality change.
35     4. The deficits we must see are start?end-symmetric. Local's  $|?start|$  ? Gemini's;
the entire gap is
36     the rally-END ( $|?end|$  2.1?3.1 s vs Gemini ~0.8?1.2 s). A single blended IoU
cannot show that ? and
37     directing R4 needs exactly that decomposition.
38
39     ---
40
41     ## 2. The metric
42
43     ### 2.1 Match rule (asymmetric tolerance ? forgive the start, expose the end)
44     A predicted window matches a golden rally iff ``|?start| ? ?_start` AND `|?end| ?
?_end`.

```

```

45     - ``?_start` generous (~2.0 s)** ? absorbs the service-window/annotator start fuzziness
(the part that is
46     *not* a model deficit).
47     - ``?_end` tighter (~1.5 s, but IAA-calibrated ? see §2.4)** ? the rally-end IS a sharp
physical event
48     (shuttle down/net/out), so a loose end *is* a real deficit we want the metric to
expose, not forgive.
49
50     > A *symmetric* ±2 s tolerance is tempting (the owner's stated bar) but the preview (§3)
shows it is *not*
51     > automatically more lenient than tIoU 0.5 and it blurs exactly the end-deficit we need
to see. Asymmetry is
52     > the point: relax the dimension that's intrinsically fuzzy (start), keep honest the one
that's a real event (end).
53
54     ### 2.2 Assignment ? strict 1:1 (Hungarian), never overlap/greedy
55     Match references to predictions by **global Hungarian assignment** (max total matches
under the tolerance
56     gate), one-to-one. This is what makes **over-production cost precision directly**: `P@?
= matched / n_pred`
57     crashes when the detector emits 75 windows for 52 rallies. (tIoU's overlap matching let
extras "match
58     something"; the W1 over-segmentation pendulum was barely costed.)
59
60     ### 2.3 Report the DECOMPOSITION, never one number
61     1. ``P@?`, `R@?`, `F1@?`** ? precision (over-production), recall (did we find the rally
within tolerance), F1.
62     2. **Boundary-error distributions** ? median +++ IQR** of signed *and* absolute
`?start`, `?end` over matched
63     pairs, **split**. This is the metric that says **"start solved, end open" and aims
R4.
64     3. **Over-seg guard (mandatory, per-video no-regression):** `split_count` (preds
covering one ref ? W1
65     over-seg) and `merge_count` (refs under one pred ? the baseline mega-window). **A
promotion may not
66     increase either on ANY video, even if mean F1 rises.**
67     4. ``?-sweep** (`? ? {1.5, 2.0, 2.5, 3.0}`, symmetric for the trend; asymmetric pair for
the headline) ?
68     because the convention/IAA noise is large, a single ? misleads (§3). Report the
curve.
69     5. ``tIoU`/`mIoU` demoted to a SECONDARY trend-line diagnostic** (keeps the paper-
ablation backbone
70     comparable) ? **not the gate.**
71
72     ### 2.4 Choosing ? ? it MUST be calibrated to inter-annotator agreement
73     The preview (§3) shows even **Gemini** scores `tolF1@1.5 = 0.20` on mp2 ? i.e. on ~half
the rallies the
74     golden END differs from Gemini's by >1.5 s. That is plausibly **label/convention noise,
not model error**:
75     we'd be measuring how two humans (or human-vs-Gemini) disagree on "when did the rally
end." **Therefore
76     `?_end` must be ? the inter-annotator end-agreement**, which we do not yet know.
**Action:** a small **IAA
77     study** ? a 2nd labeler re-labels 2?3 already-golden videos; set `?_start`/`?_end` to
~the 90th-percentile
78     pairwise human |?start|/?end|. Until then **freeze provisional ?_start=2.0, ?_end=1.5
and always report the
79     sweep**, with the `? = D.(1??)/(1+?)` derivation recorded so the choice is auditable.
80
81     ---
82
83     ## 3. Preview on the two ground-truth clips (greedy-1:1 approximation; Hungarian in the
impl)
84
85     Symmetric-? sweep, F1@? (local-best = the per-clip best config: mp2 W1+W2+hardcap,

```

```

GX010128 W1+W2):
86
87 | clip / method | tolF1@1.5 | @2.0 | @2.5 | @3.0 | (tIoU-F1@0.5) |
88 |---|---|---|---|---|---|
89 | mahadevpura-2 / **Gemini** | 0.203 | 0.354 | 0.506 | **0.557** | 0.557 |
90 | mahadevpura-2 / local | 0.111 | 0.148 | 0.222 | 0.278 | 0.296 |
91 | GX010128 / **Gemini** | 0.615 | 0.769 | 0.813 | **0.857** | 0.813 |
92 | GX010128 / local | 0.189 | 0.315 | 0.346 | 0.362 | 0.457 |
93
94 **Reading the preview (all honest):**
95 - **`tolF1@~3.0 ? tIoU-F1@0.5`** ? the sweep is a smooth, interpretable knob that
*contains* tIoU as a point;
96 good (continuity with the frozen baseline).
97 - **The local?Gemini gap WIDENS under the tighter, 1:1 tolerance** (GX010128 ratio 0.56
at tIoU0.5 ?
98 **0.41** at ?2.0). R2 charges local for the 1.5× over-production (precision) and the
loose ends ? exactly
99 the deficits tIoU under-counted. **This is the metric working, not local regressing.**
100 - **Even Gemini drops at small ?** (mp2 0.557?0.203 from ?3.0?1.5) ? strong evidence the
**golden END
101 convention carries >1.5 s of noise** ? $2.4's IAA calibration is a prerequisite, not a
nicety.
102 - Boundary-error medians (the robust diagnostic, from the validation report): local
|?start| 0.78?1.74 s
103 (? Gemini 0.98?1.46) vs local |?end| 2.12?3.12 s (? Gemini 0.76?1.19) ? **the headline
R4 signal.**
104
105 ---
106
107 ## 4. Promotion / "trustworthiness" rule (small-n)
108 A candidate is promotable only if **all** hold:
109 1. **Paired per-video sign-test sweep on F1@? is near-unanimous: **6/0 @ n=6
(p=0.031)** or **7/0 @ n=7
110 (p=0.016)** (8/0 @ n=8, p=0.008); 5/1 or 6/1 is NOT significant. Run on the
**growing-golden** set.
111 2. **No over-seg/merge-guard regression on ANY video** ($2.3.3).
112 3. **Effect size + bootstrap CI** reported as an honesty band (expect it to span 0 at
this n; use it to
113 separate "real" from "directionally-clean-but-negligible," not as the pass/fail).
114 4. **Stratify, don't blend** ? report the continuously-active blob clip (mahadevpura-1)
as its own stratum
115 (its near-degenerate scores can swing a small-n mean).
116 5. **No regression on EITHER eval set** (raw + growing-golden) ? the dual-set
discipline.
117
118 ---
119
120 ## 5. Implementation plan (pure, offline, additive ? zero production risk)
121 - **`backend/eval/tolerance_metrics.py`** (pure, GPU-free, unit-testable):
122 - `match_1to1(preds, gts, ?_start, ?_end) -> (matches, P, R, F1)` ? Hungarian via
123 `scipy.optimize.linear_sum_assignment` if available, else a pure O(n³) fallback
(corpus n is tiny).
124 - `boundary_error_report(matches) -> {?start, ?end: signed/abs median+IQR}`.
125 - `over_seg_report(preds, gts) -> {split_count, merge_count}` (reuse the bipartite
logic from
126 `segmentation_metrics.merge_split_report`).
127 - **Wire into `backend/eval/rally_seg_eval.py`** ? emit the R2 block **alongside** tIoU
(don't remove tIoU);
128 add `--tau-start/--tau-end/--tau-sweep`. Surface in the experiment leaderboard.
129 - **`run_regression.py`** ? add the dual-set R2 aggregate + the per-video over-seg-guard
gate.
130 - **Tests** ? synthetic intervals: exact-match, start-fuzzy-within-?, end-loose-
beyond-?, over-production
131 (precision crash), the Hungarian-vs-greedy tie case, empty inputs.
132 - **Default-additive:** R2 is reported next to tIoU; the *gate* switches to R2 only

```

```

after it's validated to
133     rank candidates sensibly on the golden set (so we never silently change the promotion
bar mid-flight).
134
135     ---
136
137     ## 6. Definition of done
138     - `tolerance_metrics.py` + tests merged (default-additive); `rally_seg_eval` reports the
R2 decomposition +
139     ?-sweep next to tIoU; the corpus (n?8) re-scored under R2 and recorded in
QUALITY_ITERATIONS.
140     - The IAA study (§2.4) scoped (it can land after R2's code; R2 ships with provisional ?
+ the sweep).
141     - `run_regression.py` carries the dual-set R2 gate + the per-video over-seg guard.
142     - tIoU/mIoU retained as a secondary trend-line.
143
144     ---
145
146     ## 7. Decisions
147     **Locked (recommended):** Hungarian 1:1 (not greedy/overlap); report the decomposition +
sweep, never one
148     number; tIoU demoted to secondary; R2 default-additive until validated; boundary-error
split start/end is
149     the R4-aiming diagnostic.
150     **Open ? for your deeper look:**
151     - **?_end: asymmetric 1.5 s vs symmetric 2.0 s?** (Asymmetric exposes the end-deficit ?
recommended ? but
152     needs the IAA floor.) Your "±2 s, not more" bar maps to ?_start=2.0; ?_end is the real
question.
153     - **IAA study now or after R2 code?** (Recommend: R2 code now with provisional ? +
sweep; IAA next, to fix ?.)
154     - **Replace vs augment the promotion gate** (recommend augment ? validate ? replace).
155

```

15. assistant (2026-06-17T09:05:53.001Z)

I now have the full picture. The PR is docs-only (R2 eval metric design), and §7 has three open decisions. The owner's comment:

> "I agree with both of the first recommendations but in the last one we do [aug]ment but before replacing we should run an ablation experiment and deprecate with documentation so it's reserved for posterity"

So: lock decisions 1 (asymmetric ?_end) and 2 (R2 code now, IAA after) as agreed; for decision 3, **augment** the gate, then **before replacing** tIoU run an **ablation experiment** and **deprecate tIoU with documentation** (kept for posterity, not deleted). This is reasonable and consistent with the repo's "tIoU retained as secondary trend-line" + paper-ablation-backbone norms. Let me incorporate it.

16. user (2026-06-17T09:06:05.230Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\.\claude\worktrees\docs-r2\docs\R2_EVAL_METRIC_DESIGN.md has been updated successfully. (file state is current in your context ? no need to Read it back)

17. assistant (2026-06-17T09:06:13.288Z)

Now let me make the ablation-gate + deprecation concrete in §5 (impl plan) and §6 (DoD), and update the status header.

18. user (2026-06-17T09:06:17.437Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\.\claude\worktrees\docs-r2\docs\R2_EVAL_METRIC_DESIGN.md has been updated successfully.

(file state is current in your context ? no need to Read it back)

19. user (2026-06-17T09:06:23.700Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\docs-r2\docs\R2_EVAL_METRIC_DESIGN.md has been updated successfully.
(file state is current in your context ? no need to Read it back)

20. user (2026-06-17T09:06:28.254Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\docs-r2\docs\R2_EVAL_METRIC_DESIGN.md has been updated successfully.
(file state is current in your context ? no need to Read it back)

21. assistant (2026-06-17T09:06:31.636Z)

Now let me commit in the worktree, verify HEAD is mine, push, and merge.

22. user (2026-06-17T09:06:42.313Z)

/usr/bin/bash: line 1: cd: .claude/worktrees/docs-r2: No such file or directory

23. user (2026-06-17T09:06:46.956Z)

```
docs/r2-eval-metric
--- ahead of origin/master:
38808f0 docs(eval): R2 design ? task-faithful tolerance-match rally metric (for review)
--- diff stat:
warning: in the working copy of 'docs/R2_EVAL_METRIC_DESIGN.md', LF will be replaced by CRLF the
next time Git touches it
 docs/R2_EVAL_METRIC_DESIGN.md | 42 ++++++-----
1 file changed, 29 insertions(+), 13 deletions(-)
```

24. assistant (2026-06-17T09:06:50.126Z)

Only my doc change, single file. Let me commit and push.

25. user (2026-06-17T09:07:02.085Z)

```
warning: in the working copy of 'docs/R2_EVAL_METRIC_DESIGN.md', LF will be replaced by CRLF the
next time Git touches it
[docs/r2-eval-metric bb7b9e4] docs(eval): lock R2 open decisions per owner review (2026-06-17)
1 file changed, 29 insertions(+), 13 deletions(-)
bb7b9e4 docs(eval): lock R2 open decisions per owner review (2026-06-17)
38808f0 docs(eval): R2 design ? task-faithful tolerance-match rally metric (for review)
```

26. user (2026-06-17T09:07:08.318Z)

To <https://github.com/avidullu/badminton-highlight-indexer.git>
38808f0..bb7b9e4 docs/r2-eval-metric -> docs/r2-eval-metric

27. assistant (2026-06-17T09:07:16.148Z)

Now I'll reply to the review comment and merge (docs-only, checks passing).

28. user (2026-06-17T09:07:22.427Z)

<https://github.com/avidullu/badminton-highlight-indexer/pull/198#issuecomment-4727974159>

29. user (2026-06-17T09:07:32.435Z)

failed to run git: fatal: 'master' is already used by worktree at
'C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/tier1-windowing'

30. user (2026-06-17T09:07:40.940Z)

```
{"mergeCommit":"cf8973e5d17c89ed2e5058cef195e99dcee61403","mergedAt":"2026-06-17T09:07:31Z","state":"MERGED"}
```

31. assistant (2026-06-17T09:07:47.462Z)

PR merged. The only error was gh trying to switch the local checkout to `master` (which another worktree holds) for branch cleanup. Let me clean up the branch and worktree manually.

32. user (2026-06-17T09:09:01.158Z)

--- (empty above = remote branch gone)

33. assistant (2026-06-17T09:09:06.502Z)

Remote branch was deleted. Now remove the now-orphaned worktree and prune the local branch.

34. user (2026-06-17T09:09:41.476Z)

Deleted branch docs/r2-eval-metric (was bb7b9e4).

--- remaining worktrees:

```
C:/Users/avidu/Projects/badminton-highlight-indexer
95cd154 [docs/golden-regression-fixtures-plan]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/agent-a771ff1d85e18f2ed
7e0881a [perf/wasb-fast-frames]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/bhi-execpolicy
5cb7bbf [bhi-execpolicy]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/docs-golden
c33cf1d [docs/golden-videos-tracker]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/docs-rally-quality
877ab1f [docs/rally-quality-research]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/docs-today
ce7eb47 [docs/quality-iterations-realfootage]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/gf-audit
7414446 [feat/golden-fixtures-audit-hardening]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/gf-p1
955e902 [feat/golden-fixtures-p1]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/hardchop
614325d [feat/w1-hard-cap-fallback]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/local-noai
3ea7ad4 [feat/local-no-ai-segmenter]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/r2-impl
cf8973e [feat/r2-tolerance-metric]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/tier0-eval
9e03e5a [feat/tier0-rally-seg-eval]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/tier1-windowing
a3b209a [master]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/w2-netcross
effd920 [feat/w2-netcross-gate]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/w2-safeguard
5ba6955 [feat/w2-never-empty-safeguard]
```

35. assistant (2026-06-17T09:09:53.099Z)

Done.

Summary

[PR #198](https://github.com/avidullu/badminton-highlight-indexer/pull/198) (`docs(eval): R2 design ? task-faithful tolerance-match rally metric`) ? incorporated the owner review comment and **merged** (squash, commit `cf8973e`; branch deleted, orphaned worktree cleaned up).

The review comment approved decisions 1 & 2 and asked, on decision 3, to **augment** now and ?

before replacing tIoU ? run an ablation experiment and deprecate it with documentation "so it's reserved for posterity."

****What I changed**** in docs/R2_EVAL_METRIC_DESIGN.md, moving the three §7 open decisions to *locked (owner review, 2026-06-17)*:

1. ****Asymmetric ?_end (1.5 s) / ?_start (2.0 s) ? adopted.****
2. ****IAA study after R2 code ? adopted**** (ship with provisional ? + sweep).
3. ****Gate ? augment ? ablation-gate the replacement ? deprecate-with-docs:**** R2 reported alongside tIoU until a dedicated ablation experiment validates it ranks candidates at least as sensibly; when tIoU is retired as the gate it's formally deprecated *with documentation, never deleted* (stays the secondary trend-line + `QUALITY\_ITERATIONS.md` paper-ablation backbone, with a dated note + ablation evidence).

Also reflected the gate flow in §5 (impl plan), §6 (DoD), and the status header (now "owner-reviewed, decisions locked 2026-06-17"). I posted a reply comment on the PR documenting the incorporation before merging.

Notes: this was docs-only with CI green, so it fit the merge authorization for verified-green non-product PRs; ****R2 implementation itself remains owner-gated and not started.**** I worked in the existing `docs-r2` worktree, branched from the up-to-date remote, staged only the one doc file, and verified HEAD held only my commits before pushing.

Want me to refresh the current-state/handoff memory to note this decision is locked?